Bowel Gas as an Acoustic Transducer: A Constrained Bubble Model for Infrasound-Induced Mechanotransduction in the Gastrointestinal Tract

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Abstract

Whole-body acoustic models predict negligible tissue displacement from airborne infrasound, yet gastrointestinal (GI) discomfort is anecdotally reported at high sound pressure levels (SPL). We propose that *intraluminal gas pockets* act as impedance-matching transducers, converting pressure oscillations to local wall strain far more efficiently than whole-cavity resonance. A constrained-bubble model—Minnaert dynamics modified for the elastic intestinal wall and cylindrical lumen geometry—shows that gas pockets of 5 mL to 100 mL have resonant frequencies from tens to hundreds of hertz, depending on geometry, but sub-resonantly amplify tissue displacement as compliant inclusions in an otherwise stiff medium. The SPL threshold for activating Piezo mechanosensitive ion channels ranges from 111 dB (100 mL pocket) to 120 dB (5 mL). A Monte Carlo population model ($N = 10\,000$) predicts that nearly all simulated individuals exceed the Piezo threshold at 120 dB, a consequence of the fact that every individual harbours at least several millilitres of bowel gas. This pathway is 35–100× more efficient than airborne coupling to abdominal flexural modes and represents the most plausible acoustic mechanism identified to date for airborne-infrasound-induced GI mechanotransduction, with enhanced susceptibility in individuals with elevated bowel gas.

Keywords: infrasound, bowel gas, constrained bubble, mechanotransduction, Piezo channels, gastrointestinal acoustics, Minnaert resonance

1 Introduction

The so-called "brown note"—a hypothetical infrasonic frequency that induces involuntary bowel movements—has persisted in popular culture and military speculation for decades, despite a lack of credible experimental evidence [Jauchem and Cook, 2007, Leventhall, 2007,

Parker and Stone, 2000, Savage and Hyneman, 2005]. Early aerospace-medicine studies documented nausea, abdominal vibration, and vague GI discomfort in subjects exposed to high-intensity infrasound (>140 dB), but attributed these effects primarily to whole-body vibration rather than airborne acoustic coupling [Mohr et al., 1965, von Gierke, 1974].

Our companion study [Mace et al., 2026] demonstrated that airborne infrasound cannot excite abdominal-cavity flexural resonances at physiologically meaningful amplitudes. The breathing mode of the abdomen—often conflated with the brown-note myth in popular accounts—lies at ∼2490 Hz, far above the infrasound range. The relevant flexural modes ($n \geq 2$) do fall at 4 Hz to 10 Hz, but the $(ka)^n$ coupling penalty renders airborne excitation negligible: the required SPL for 1 µm tissue displacement exceeds 180 dB.

However, the whole-cavity analysis treats the abdomen as a homogeneous elastic shell filled with incompressible fluid. This neglects a critical feature: the human gastrointestinal tract normally contains 100 mL to 200 mL of gas distributed in multiple discrete pockets [Levitt, 1971, Suarez et al., 1997], with pathological or post-prandial values reaching 500 mL to 1500 mL [Bedell et al., 1956, Serra et al., 2001]. These gas pockets constitute *compressible inclusions* within an otherwise nearly incompressible medium, creating local impedance mismatches of ∼ 3600:1 at each gas–tissue interface.

We propose that bowel gas pockets function as *acoustic transducers*: the spatially uniform infrasonic pressure field drives radial oscillation of each gas–tissue interface, converting acoustic pressure to local wall strain. This mechanism is fundamentally different from—and far more efficient than—whole-cavity resonance, because:

1. Gas compressibility provides a compliance that tissue lacks;
2. The transduction is broadband (not limited to a single resonance);
3. The displacement is concentrated at the gas–tissue boundary, where mechanosensitive ion channels (Piezo1, Piezo2) are expressed in enterochromaffin cells and enteric neurons [Coste et al., 2010, Alcaino et al., 2017, Zeitzschel and Lechner, 2024].

In this paper we develop a constrained-bubble model for intraluminal gas pockets, compute SPL thresholds for Piezo channel activation, quantify the dependence on pocket volume and geometry, and use Monte Carlo simulation to predict population-level variability.

2 Model

2.1 Geometry and assumptions

We model a single gas pocket as a compressible inclusion within the intestinal lumen, surrounded by an elastic wall of thickness h_w, Young's modulus E_w, Poisson's ratio ν_w, and density ρ_w (Fig. 1). The surrounding tissue is modelled as an incompressible fluid of density ρ_f at infrasound wavelengths ($\lambda \approx 50$ m at 7 Hz, $\gg$ body dimension ∼0.3 m).

Two idealized geometries bracket the range of real pocket shapes:

- **Spherical:** radius $a = (3V/4\pi)^{1/3}$, appropriate for isolated gas pockets in distended segments.

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- **Cylindrical:** fixed lumen radius $R = 15\,\mathrm{mm}$ (small intestine), length $L = V/(\pi R^2)$, appropriate for elongated gas slugs.

Canonical parameter values are listed in Table 1.

Table 1: Model parameters.

Symbol	Description	Value	Source
γ	Specific-heat ratio (air)	1.4	—
P_0	Ambient (atmospheric) pressure	101 325 Pa	—
ρ_f	Tissue/fluid density	$1020\,\mathrm{kg/m^3}$	Fung [1993]
E_w	Intestinal wall Young's modulus	10 kPa	Gregersen and Kassab [2000]
ν_w	Intestinal wall Poisson's ratio	0.45	Fung [1993]
h_w	Intestinal wall thickness	3 mm	Higa et al. [2007]
ρ_w	Wall density	$1040\,\mathrm{kg/m^3}$	—
c_f	Longitudinal wave speed in tissue	1540 m/s	—
μ	Dynamic viscosity	$1 \times 10^{-3}\,\mathrm{Pa \cdot s}$	—
R	Lumen radius (small intestine)	15 mm	—

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2.2 Resonance frequency

2.2.1 Spherical pocket: modified Minnaert frequency

The classical Minnaert frequency for a free spherical gas bubble of radius a in an infinite liquid is [Minnaert, 1933]:

$$f_M = \frac{1}{2\pi a}\sqrt{\frac{3\gamma P_0}{\rho_f}}. \tag{1}$$

For a pocket enclosed by an elastic shell (the intestinal wall), both the gas stiffness and the wall stiffness contribute, while the inertia includes both the radiation added mass of the surrounding fluid and the shell mass [Church, 1995, Hoff et al., 2000]. Expressing all quantities as stiffness and inertia per unit radial displacement:

$$f_0^{(\mathrm{sph})} = \frac{1}{2\pi}\sqrt{\frac{k_{\mathrm{gas}} + k_{\mathrm{wall}}}{m_{\mathrm{fluid}} + m_{\mathrm{wall}}}}, \tag{2}$$

where

$$k_{\mathrm{gas}} = \frac{3\gamma P_0}{a}, \qquad k_{\mathrm{wall}} = \frac{2\,E_w\,h_w}{a^2\,(1-\nu_w)}, \tag{3}$$

$$m_{\mathrm{fluid}} = \rho_f\,a, \qquad m_{\mathrm{wall}} = \rho_w\,h_w. \tag{4}$$

The gas stiffness arises from adiabatic compression of the enclosed gas (Boyle's law linearised about equilibrium radius a); the wall stiffness from hoop stress in a thin spherical shell; and the inertia terms from radiation loading and shell mass per unit area, respectively. In the

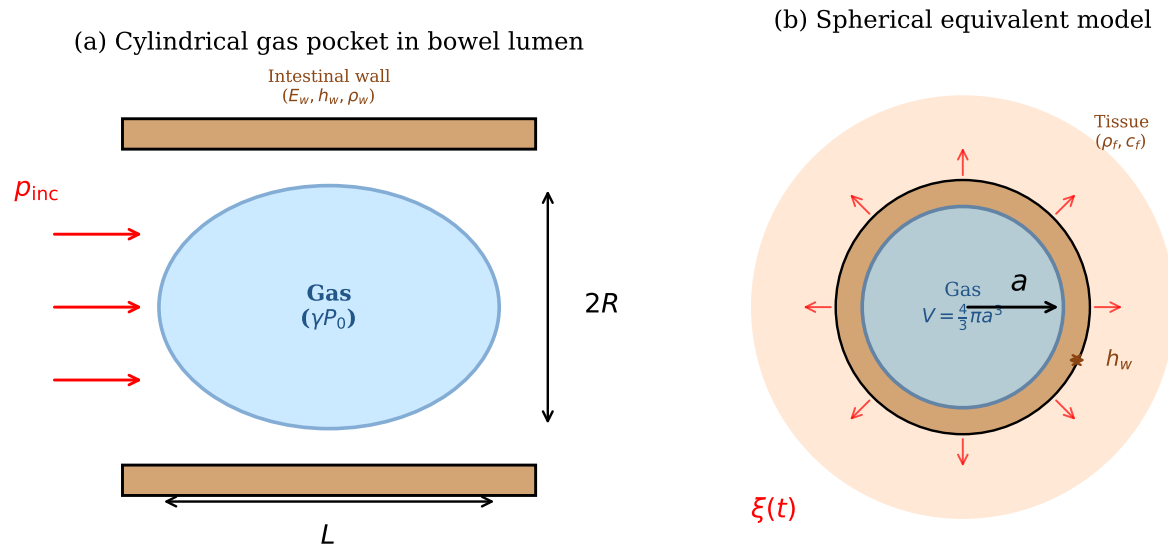

Figure 1: (a) Cylindrical gas pocket constrained within the intestinal lumen. The incident pressure p_{inc} is spatially uniform at infrasound wavelengths. (b) Spherical equivalent model showing radial oscillation $\xi(t)$ of the gas–tissue interface, elastic wall of thickness h_w, and surrounding tissue of density ρ_f.

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limit $h_w \to 0$ (free bubble), k_{wall} and m_{wall} vanish and Eq. (2) reduces exactly to the Minnaert result, Eq. (1):

$$f_0 \to \frac{1}{2\pi}\sqrt{\frac{3\gamma P_0/a}{\rho_f\, a}} = \frac{1}{2\pi a}\sqrt{\frac{3\gamma P_0}{\rho_f}} = f_M.$$

For the soft intestinal wall ($E_w = 10\,\text{kPa}$), k_{wall} contributes only 0.9–2.4% of the total effective stiffness $k_{\text{gas}} + k_{\text{wall}}$ across the physiological pocket-size range; gas compressibility dominates. The wall mass $m_{\text{wall}} = \rho_w h_w$ adds ∼20% to the fluid inertia, which lowers f_0 slightly below f_M. The "constrained" label therefore refers to the geometric confinement of the gas pocket within the tubular lumen, not to a significant elastic restraint by the wall.

2.2.2 Cylindrical pocket: radial and axial modes

For an elongated gas column in an elastic tube of radius R, the lowest *radial* (breathing) mode has frequency:

$$f_0^{(\text{cyl,rad})} = \frac{1}{2\pi}\sqrt{\frac{2\gamma P_0/R + E_w h_w/[R^2(1-\nu_w^2)]}{\rho_f R + \rho_w h_w}}. \tag{5}$$

The factor of 2 (vs. 3 for the sphere) reflects the cylindrical geometry.

The *axial* (piston) mode treats the gas slug as a spring with stiffness $k = \gamma P_0 \pi R^2/L$ and radiation mass $m_{\text{end}} = \frac{8}{3}\rho_f R^3$ at each open end [Leighton, 1994]:

$$f_0^{(\text{cyl,ax})} = \frac{1}{2\pi}\sqrt{\frac{\gamma P_0\, \pi R^2}{2L \cdot \frac{8}{3}\rho_f R^3}}. \tag{6}$$

The lowest resonance for a cylindrical pocket is $f_0 = \min(f_0^{(\mathrm{cyl,rad})}, f_0^{(\mathrm{cyl,ax})})$.

For physiological gas pocket volumes (1 mL to 100 mL), resonance frequencies range from ∼30 Hz (large cylindrical pockets, axial mode) to ∼440 Hz (small spherical pockets with elastic wall). Because wall stiffness adds little and wall mass lowers f_0 below the free-bubble Minnaert value, the constrained spherical frequencies are roughly 89–96% of f_M for clinically relevant volumes of 5 mL to 100 mL (Table 2). All resonances remain well above 20 Hz—i.e. above the conventional infrasound boundary. This is a critical distinction from earlier analyses that sought resonance *at* infrasound frequencies.

2.3 Damping

The total damping ratio is

$$\zeta = \tfrac{1}{2}(\delta_{\mathrm{rad}} + \delta_{\mathrm{th}} + \delta_{\mathrm{vis}} + \delta_{\mathrm{wall}}), \tag{7}$$

with contributions from:

$$\delta_{\mathrm{rad}} = \frac{\omega a}{c_f} \qquad \text{(radiation into surrounding tissue)}, \tag{8}$$

$$\delta_{\mathrm{th}} \approx 0.04 \qquad \text{(thermal conduction in gas)}, \tag{9}$$

$$\delta_{\mathrm{vis}} = \frac{4\mu}{\rho_f\, \omega\, a^2} \qquad \text{(viscous boundary layer)}, \tag{10}$$

$$\delta_{\mathrm{wall}} \approx 0.3 \qquad \text{(structural loss in intestinal wall [Fung, 1993])}. \tag{11}$$

2.4 Forced response: sub-resonant transduction

The key mechanism is not *resonance* but *compliance*: gas pockets respond to the incident pressure because they are compressible, even far below their resonant frequency. The radial displacement of the gas–tissue interface under a spatially uniform incident pressure p_{inc} is:

$$\xi = \frac{p_{\mathrm{inc}}}{k_{\mathrm{eff}}} \cdot H(\omega), \tag{12}$$

where k_{eff} is the effective radial stiffness (gas plus wall contributions) and $H(\omega)$ is the standard single-degree-of-freedom frequency response:

$$H(\omega) = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{\omega}{\omega_0} = \frac{f}{f_0}. \tag{13}$$

In the sub-resonant regime ($f \ll f_0$, so $r \ll 1$), $H \to 1$ and the displacement becomes frequency-independent:

$$\xi_{\text{sub-res}} \approx \frac{p_{\mathrm{inc}}}{k_{\mathrm{eff}}}. \tag{14}$$

For a spherical pocket with elastic wall:

$$k_{\mathrm{eff}}^{(\mathrm{sph})} = \frac{3\gamma P_0}{a} + \frac{2\, E_w\, h_w}{a^2(1-\nu_w)}. \tag{15}$$

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For a cylindrical pocket:

$$k_{\text{eff}}^{(\text{cyl})} = \frac{2\gamma P_0}{R} + \frac{E_w\, h_w}{R^2(1-\nu_w^2)}. \tag{16}$$

The incident pressure at SPL S (in dB re 20 µPa) is:

$$p_{\text{inc}} = p_{\text{ref}} \times 10^{S/20}, \qquad p_{\text{ref}} = 20\,\mu\text{Pa}. \tag{17}$$

2.5 Volume displacement and tissue coupling

The oscillating gas–tissue interface radiates displacement into the surrounding tissue as a monopole source. The volume displacement is:

$$\Delta V = \begin{cases} 4\pi a^2\,\xi & \text{(spherical)}, \\ 2\pi R\,L\,\xi & \text{(cylindrical, radial mode)}. \end{cases} \tag{18}$$

In the near field, the tissue displacement at distance r from the pocket centre decays as:

$$u(r) \approx \left(\frac{a}{r}\right)^2 \xi, \qquad r > a. \tag{19}$$

This concentrates the acoustic displacement at the gas–tissue boundary, precisely where Piezo channels are located in the intestinal wall [Alcaino et al., 2017, Mazzuoli-Weber and Schemann, 2015].

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2.6 PIEZO activation criterion

We adopt a displacement threshold of $\xi_{\text{PIEZO}} = 0.5\,\mu\text{m}$ for activation of Piezo2 mechanosensitive ion channels, based on patch-clamp measurements of membrane indentation required for channel gating [Coste et al., 2010]. This is a conservative (most sensitive) value; typical activation requires 0.5 µm to 2.0 µm.

The minimum SPL for Piezo activation is found by solving $\xi(\text{SPL}) = \xi_{\text{PIEZO}}$:

$$\text{SPL}_{\text{thresh}} = 20\log_{10}\left(\frac{\xi_{\text{PIEZO}}\, k_{\text{eff}}}{p_{\text{ref}}\, H(\omega)}\right). \tag{20}$$

2.7 Population variability: Monte Carlo model

Inter-individual variability in bowel gas content is substantial [Levitt, 1971, Bedell et al., 1956, Serra et al., 2001]. We model a population of $N = 10\,000$ individuals with:

- Total bowel gas volume drawn from a log-normal distribution with median 200 mL and $\sigma_{\ln} = 0.65$, giving a 95% CI of approximately 56 mL to 715 mL and a heavy right tail extending to ∼1500 mL;
- Number of discrete pockets $N_p \sim \text{Poisson}(\lambda = 10)$, clipped to [2, 40];
- Per-pocket volumes drawn from a symmetric Dirichlet partition of the total volume;

- Geometry randomly assigned (70% cylindrical, 30% spherical).

For each individual, the maximum pocket-wall displacement at 7 Hz and 120 dB is computed, and the fraction exceeding ξ_{PIEZO} is recorded.

3 Results

3.1 Resonance frequencies are above infrasound

Table 2 summarizes the resonance frequencies for representative gas pocket sizes. Spherical pocket resonances range from approximately 108 Hz at 100 mL to 438 Hz at 1 mL, and the cylindrical radial (breathing) mode lies at ∼162 Hz. However, cylindrical *axial* (piston) modes fall below 100 Hz for pockets larger than ∼10 mL, reaching 31 Hz at 100 mL—near, but still above, the conventional infrasound boundary (20 Hz). None of the pockets resonate *at* infrasound frequencies; the sub-resonant compliance mechanism described below operates efficiently across the entire infrasound band.

Table 2: Resonance frequencies for elastic-wall gas pockets.

V [mL]	Geometry	a or R [mm]	f_{Minnaert} [Hz]	f_0 [Hz]
1	Spherical	6.2	524	438
5	Spherical	10.6	306	273
10	Spherical	13.4	243	222
20	Spherical	16.8	193	179
50	Spherical	22.9	142	134
100	Spherical	28.8	113	108
10	Cylindrical†	15.0	—	99
50	Cylindrical†	15.0	—	44
100	Cylindrical†	15.0	—	31

†Cylindrical f_0 is the axial (piston) mode, which is lowest. The radial (breathing) mode is ∼162 Hz, independent of volume.

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3.2 SPL thresholds for PIEZO activation

Figure 2 shows the SPL required to produce 0.5 µm wall displacement as a function of gas pocket volume. Key findings:

- For **spherical** pockets, the threshold decreases from ∼120 dB at 5 mL to ∼111 dB at 100 mL.
- For **cylindrical** pockets, the threshold is nearly independent of volume (∼114 dB), because the lumen radius R is fixed.
- The 9-dB range across physiological pocket sizes maps directly to the range of inter-individual gas content.

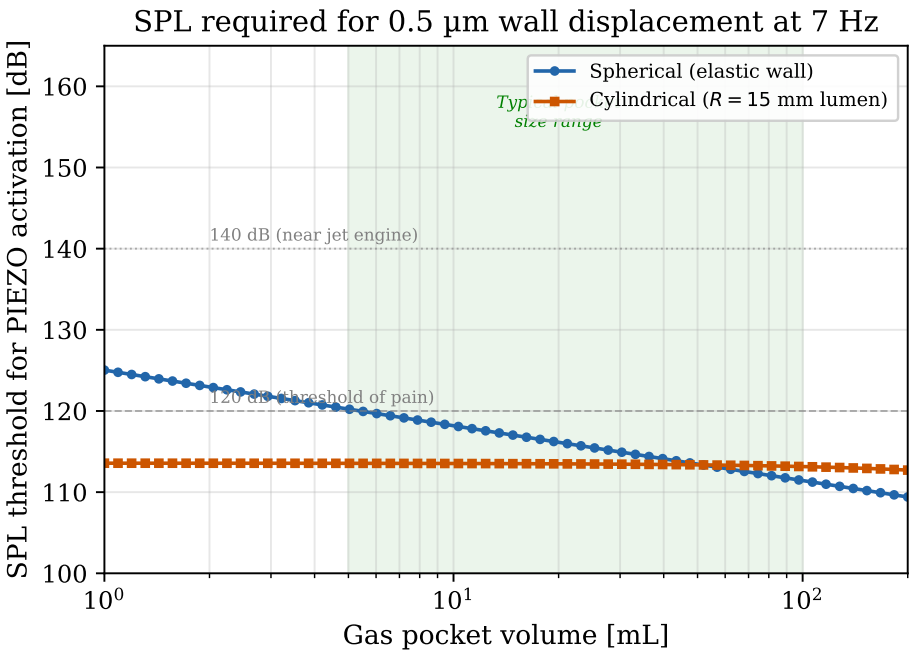

Figure 2: SPL threshold for PIEZO activation (0.5 µm wall displacement) as a function of gas pocket volume at 7 Hz. The shaded region indicates the typical physiological pocket size range. Horizontal lines mark 120 dB (threshold of pain) and 140 dB (near jet engine).

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3.3 Population variability

Figure 3 presents the Monte Carlo results for $N = 10\,000$ simulated individuals at 120 dB. The key result is that **nearly all simulated individuals exceed the PIEZO activation threshold** at this SPL. This high exceedance fraction is expected rather than surprising: the model assigns every individual at least 30 mL of bowel gas (a physiologically conservative lower bound), and even a 3 mL cylindrical pocket produces $\xi > 0.5$ µm at 120 dB. The result therefore confirms internal consistency of the model rather than constituting an independent prediction; experimental validation of the threshold remains necessary (see §4.5).

The distribution of maximum pocket-wall displacement spans approximately 1.03 µm to 2.70 µm, with a median of ∼1.08 µm. Individuals in the upper tail (>500 mL total gas) experience displacements exceeding 2 µm, well into the range of robust PIEZO activation.

3.4 Comparison with whole-cavity resonance

Figure 4 compares the gas-pocket transduction pathway with the whole-cavity flexural mode ($n = 2$) pathway from Mace et al. [2026]. The gas-pocket mechanism produces tissue displacement that is **35–100× larger** than the whole-cavity pathway at the same SPL, across the entire range from 80 dB to 150 dB.

At 120 dB, the whole-cavity $n = 2$ mode yields $\xi \approx 0.014$ µm (energy-consistent; Mace et al., 2026)—more than an order of magnitude below the PIEZO threshold. In contrast, a 100 mL gas pocket yields $\xi \approx 1.3$ µm, comfortably above threshold.

4 Discussion

4.1 Gas pockets as impedance-matching transducers

The central finding of this paper is that intraluminal gas pockets serve as *impedance-matching acoustic transducers* between the airborne pressure field and the intestinal wall. The mechanism is not resonance enhancement—the gas-pocket resonances still lie well above the 7 Hz driving frequency, at roughly 31 Hz to 438 Hz across the cases studied—but rather the *compliance mismatch* between the compressible gas and the surrounding incompressible tissue.

At infrasound wavelengths ($\lambda \gg$ body size), the incident pressure is spatially uniform throughout the abdomen. In incompressible tissue, this uniform pressure produces negligible strain (bulk modulus ~2.2 GPa). But at a gas–tissue interface, the gas compresses while the tissue does not, creating a displacement discontinuity that scales as $\xi \propto p_{\text{inc}}/k_{\text{eff}}$. This is fundamentally the same mechanism by which the middle ear's ossicular chain provides impedance matching between air and the cochlear fluid [Junger and Feit, 1986], albeit at much lower frequencies and without mechanical advantage.

4.2 Acoustic short-circuit via the GI air column

An important caveat concerns the acoustic pathway through luminal air. The GI tract is an open tube (mouth to anus, total length ~8 m). At infrasound wavelengths ($\lambda = c_{\text{air}}/f \approx 49$ m at 7 Hz), $\lambda \gg L_{\text{GI}}$, so luminal air could in principle deliver the same incident pressure p_{inc} to the *interior* of a gas pocket as tissue delivers to its *exterior*. If both paths transmit p_{inc} faithfully, the net pressure difference across the gas–tissue interface vanishes: $\Delta P \to 0$, and no displacement results—an *acoustic short circuit*.

We estimate the cross-over frequency by modelling the open GI tract as a Helmholtz resonator. The luminal gas volume V communicates with the exterior through a constricted air column of cross-section S and effective acoustic length L_{eff}. The Helmholtz resonance frequency is

$$f_H = \frac{c_{\text{air}}}{2\pi}\sqrt{\frac{S}{V\,L_{\text{eff}}}}. \tag{21}$$

For representative values ($V \approx 200$ mL, $S \approx \pi(5\,\text{mm})^2$, $L_{\text{eff}} \approx 5$ m), this gives $f_H \approx 15$ Hz. Here L_{eff} represents the effective acoustic path length from mouth to colon (~5 m of GI tract), and $d = 10$ mm is a conservative lower bound for the average intestinal lumen diameter (typical range 10 mm to 25 mm). Below f_H, the air column can quasi-statically equalize pressure, partially short-circuiting the transduction mechanism; above f_H, air-column inertia prevents equalization and the constrained-bubble model applies without qualification. We note that Eq. (21) is used here only to estimate f_H as a cross-over scale; the implied low-pass attenuation model $1/[1 + (f/f_H)^2]$ is a simplification. A true Helmholtz resonator exhibits amplification near f_H, which would complicate the short-circuit picture further and warrants future investigation.

Three observations mitigate this concern at typical infrasound frequencies (f ~1 Hz to 20 Hz):

1. **Liquid and fecal plugs seal GI segments.** The intestinal lumen is not a continuous air column. Chyme, mucus, and fecal matter create liquid plugs that partition the

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tract into acoustically isolated compartments. Imaging studies confirm that gas is distributed in discrete, separated pockets rather than a single connected volume [Levitt, 1971, Serra et al., 2001]. Within each sealed segment, the luminal air path is blocked and the constrained-bubble model is fully valid.

2. **Segmented gas distribution is the norm.** Even in healthy fasting individuals, bowel gas typically occupies 5–15 discrete pockets separated by liquid or collapsed lumen [Suarez et al., 1997]. Post-prandially, the degree of segmentation increases further. Complete continuity from mouth to anus via an uninterrupted air column is physiologically exceptional.

3. **Partial equalization reduces but does not eliminate transduction.** At 7 Hz ($< f_H$), the short-circuit transfer function provides partial equalization (estimated pressure-equalization ratio $\sim$0.8 for an open segment). Quantitatively, however, the margin above the PIEZO threshold is modest: even for the largest pocket considered (100 mL, spherical), the maximum survivable reduction factor is only $\sim$2.7$\times$ before the displacement falls below 0.5 µm. A factor-of-three reduction in effective ΔP would therefore bring all pockets below the mechanotransduction threshold at 120 dB. This underscores the importance of the sealed-pocket defence (items 1–2 above): for gas pockets isolated by fecal or liquid plugs—the physiologically typical configuration—equalization cannot occur and the constrained-bubble model applies in full.

In summary, the acoustic short-circuit effect is real in principle but largely negated *in vivo* by the segmented nature of bowel gas distribution. Future work should quantify the acoustic transmission loss through liquid plugs of varying length and viscosity.

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4.3 Clinical implications

The model predicts that individuals with elevated bowel gas are more susceptible to infrasound-induced GI effects. Several clinical populations have chronically elevated intestinal gas:

- **Irritable bowel syndrome (IBS):** Patients with IBS exhibit altered gas handling and increased gas retention [Serra et al., 2001], as well as visceral hypersensitivity that may lower the effective PIEZO threshold.
- **Post-prandial states:** Gas production increases significantly after meals, particularly with fermentable substrates, transiently increasing both total gas volume and the number of discrete pockets [Harder et al., 2003].
- **Aerophagia:** Excessive air swallowing introduces large gas volumes into the upper GI tract.
- **Small intestinal bacterial overgrowth (SIBO):** Excess fermentation produces elevated luminal gas.

This framework naturally explains the sporadic and individually variable nature of reported infrasound effects: susceptibility depends on a physiological parameter (gas content) that varies by up to 15$\times$ across the population and fluctuates within individuals on hourly timescales.

4.4 Comparison with established bubble dynamics

Our constrained-bubble model builds on the extensive literature on encapsulated microbubbles used as ultrasound contrast agents [Church, 1995, Hoff et al., 2000]. The key differences are:

1. **Scale:** Ultrasound contrast agents are ∼1 µm to 10 µm in diameter; bowel gas pockets are 1 mm to 60 mm.

2. **Wall properties:** Contrast agent shells are polymeric ($E \sim 1\,\mathrm{GPa}$); the intestinal wall is a soft biological tissue ($E \sim 10\,\mathrm{kPa}$).

3. **Frequency regime:** Contrast agents are driven near resonance (1 MHz to 10 MHz); bowel gas pockets are driven far below resonance ($f/f_0 \sim 0.01$–0.05).

Despite these differences, the underlying physics is identical: pressure-driven radial oscillation of a compressible inclusion in an elastic medium. The sub-resonant regime simply replaces the resonance-amplified response with a quasi-static compliance-dominated response.

4.5 Limitations

Several simplifications warrant discussion:

1. **Linear model:** We assume small-amplitude oscillations. At very high SPL (> 140 dB), nonlinear effects (rectified diffusion, microstreaming) may become relevant.

2. **Idealized geometries:** Real gas pockets have irregular shapes. Our spherical and cylindrical models bracket the physiological range but do not capture complex topologies. The cylindrical breathing-mode model assumes $L/R \geq 1$; for pockets smaller than ∼10 mL in a 15 mm lumen, the spherical model is more appropriate as the pocket geometry becomes disc-like.

3. **Quasi-static tissue response:** The monopole coupling model (Eq. 19) neglects viscoelastic relaxation, which would reduce the displacement at longer distances.

4. **Threshold uncertainty:** The Piezo activation threshold of 0.5 µm is based on *in vitro* patch-clamp data; the *in vivo* threshold in the intestinal wall may differ due to cell morphology and mechanical pre-stress.

5. **No experimental validation:** The model predictions remain to be tested experimentally. An *in vitro* setup using a gas-filled elastic tube in a water tank with infrasonic excitation would provide direct validation.

6. **Acoustic short circuit:** As discussed in §4.2, the GI air column can partially equalize pressure across the gas–tissue interface at frequencies below $f_H \approx 15\,\mathrm{Hz}$. The model is strictly valid only for gas pockets sealed by liquid or fecal plugs—which imaging evidence suggests is the typical *in vivo* configuration—but the degree of acoustic isolation in any given individual is unknown.

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5 Conclusion

We have shown that intraluminal gas pockets in the gastrointestinal tract act as acoustic transducers, converting airborne infrasonic pressure to local wall strain via a sub-resonant compliance mechanism. This constrained-bubble pathway is 35–100× more efficient than whole-cavity flexural resonance and is the most plausible acoustic mechanism identified to date by which airborne infrasound at realistic exposure levels ($\leq$120 dB) could activate mechanosensitive ion channels in the intestinal wall.

The model makes several testable predictions:

1. SPL thresholds for GI effects should correlate with bowel gas content (measurable by abdominal X-ray or CT);
2. Post-prandial subjects should be more susceptible than fasting subjects;
3. Simethicone (which reduces gas bubble size) should attenuate effects;
4. The effect should be broadband (not frequency-specific), with magnitude scaling as $p_{\mathrm{inc}}/k_{\mathrm{eff}}$.

These predictions are amenable to experimental testing and may inform occupational exposure guidelines for environments with high-intensity infrasound.

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Data availability

All computational code is available at https://github.com/JonathanMace/brownnote.

Declaration of competing interest

The authors declare no competing interests.

References

Constanza Alcaino, Gianrico Farrugia, and Purna C. Bhattarai. Mechanosensitive ion channel Piezo2 is expressed in enteric neurons and enterochromaffin cells. *Journal of Comparative Neurology*, 525(2):400–410, 2017.

George N. Bedell, Robert Marshall, Arthur B. DuBois, and James H. Harris. Measurement of the volume of gas in the gastrointestinal tract. values in normal subjects and ambulatory patients. *Journal of Clinical Investigation*, 35(3):336–345, 1956. doi: 10.1172/JCI103284.

Charles C. Church. The effects of an elastic solid surface layer on the radial pulsations of gas bubbles. *Journal of the Acoustical Society of America*, 97(3):1510–1521, 1995. doi: 10.1121/1.412091.

Bertrand Coste, Jayanti Mathur, Michael Schmidt, Tong J. Earley, Sanjeev Ranade, Matt J. Petrus, Adrienne E. Dubin, and Ardem Patapoutian. Piezo1 and piezo2 are essential components of distinct mechanically activated cation channels. *Science*, 330(6000):55–60, 2010. doi: 10.1126/science.1193270.

Yuan-Cheng Fung. *Biomechanics: Mechanical Properties of Living Tissues.* Springer, 2nd edition, 1993. doi: 10.1007/978-1-4757-2257-4.

Hans Gregersen and Ghassan S. Kassab. Biomechanics of the gastrointestinal tract. *Neurogastroenterology and Motility*, 8(4):277–297, 2000. doi: 10.1007/BF02088063.

H. Harder, Jordi Serra, Fernando Azpiroz, M. C. Passos, S. Aguadé, and Juan-R Malagelada. Intestinal gas distribution determines abdominal symptoms. *Gut*, 52(12):1708–1713, 2003. doi: 10.1136/gut.52.12.1708.

Masato Higa, Yong Luo, Tadashi Okuyama, Yoshinori Shiraishi, Hongbin Liu, Tomoyuki Yambe, and Toshiyuki Takagi. In vivo measurements and constitutive modeling of colon tissue. *Lecture Notes in Computer Science*, 4791:285–292, 2007.

Lars Hoff, Per Christian Sontum, and Jens M. Hovem. Oscillations of polymeric microbubbles: Effect of the encapsulating shell. *Journal of the Acoustical Society of America*, 107 (4):2272–2280, 2000. doi: 10.1121/1.428557.

James R. Jauchem and Michael C. Cook. High-intensity acoustics for military nonlethal applications: A lack of useful systems. *Military Medicine*, 172(2):182–189, 2007. doi: 10.7205/MILMED.172.2.182.

Miguel C. Junger and David Feit. *Sound, Structures, and Their Interaction.* MIT Press, 2nd edition, 1986.

Timothy G. Leighton. *The Acoustic Bubble.* Academic Press, 1994. ISBN 978-0-12-441920-9.

Geoff Leventhall. What is infrasound? *Progress in Biophysics and Molecular Biology*, 93 (1–3):130–137, 2007. doi: 10.1016/j.pbiomolbio.2006.07.006.

Michael D. Levitt. Volume and composition of human intestinal gas determined by means of an intestinal washout technic. *New England Journal of Medicine*, 284(25):1394–1398, 1971. doi: 10.1056/NEJM197106242842502.

Jonathan Mace, Brian R. Mace, and GitHub Copilot CLI. Modal analysis of a fluid-filled viscoelastic oblate spheroidal shell: Implications for the existence of the "brown note". *Journal of Sound and Vibration*, 2026. Submitted.

Gemma Mazzuoli-Weber and Michael Schemann. Mechanosensitive enteric neurons in the myenteric plexus of the mouse intestine. *PLoS ONE*, 10(8):e0135828, 2015. doi: 10.1371/journal.pone.0135828.

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Marcel Minnaert. On musical air-bubbles and the sounds of running water. *Philosophical Magazine*, 16(104):235–248, 1933. doi: 10.1080/14786443309462277.

G. C. Mohr, J. N. Cole, E. Guild, and H. E. von Gierke. Effects of low frequency and infrasonic noise on man. *Aerospace Medicine*, 36(9):817–824, 1965.

Trey Parker and Matt Stone. World wide recorder concert ("the brown noise"). *South Park*, Season 3, Episode 17, first aired 12 January 2000 on Comedy Central, 2000. URL https://en.wikipedia.org/wiki/The_Brown_Noise. Directed by Eric Stough; production code 317.

Adam Savage and Jamie Hyneman. Brown note. *MythBusters*, Season 3, Episode 1 (overall Episode 25), first aired 16 February 2005 on Discovery Channel, 2005. URL https://mythresults.com/episode25. Brown-note myth tested and busted.

Jordi Serra, Fernando Azpiroz, and Juan-R Malagelada. Intestinal gas dynamics and tolerance in humans. *Gastroenterology*, 120(1):218–226, 2001. doi: 10.1053/gast.2001.20874.

Fabrizis L. Suarez, Julie K. Furne, John Springfield, and Michael D. Levitt. Insights into human colonic physiology obtained from the study of flatus composition. *American Journal of Physiology — Gastrointestinal and Liver Physiology*, 272(5):G1028–G1033, 1997. doi: 10.1152/ajpgi.1997.272.5.G1028.

Henning E. von Gierke. Effects of infrasound on man. Technical Note TN D-7620, NASA, 1974.

Nadine Zeitzschel and Stefan G. Lechner. Mechanosensitive ion channels in gastrointestinal mechanotransduction. *Channels*, 18(1):2355123, 2024. doi: 10.1080/19336950.2024.2355123.

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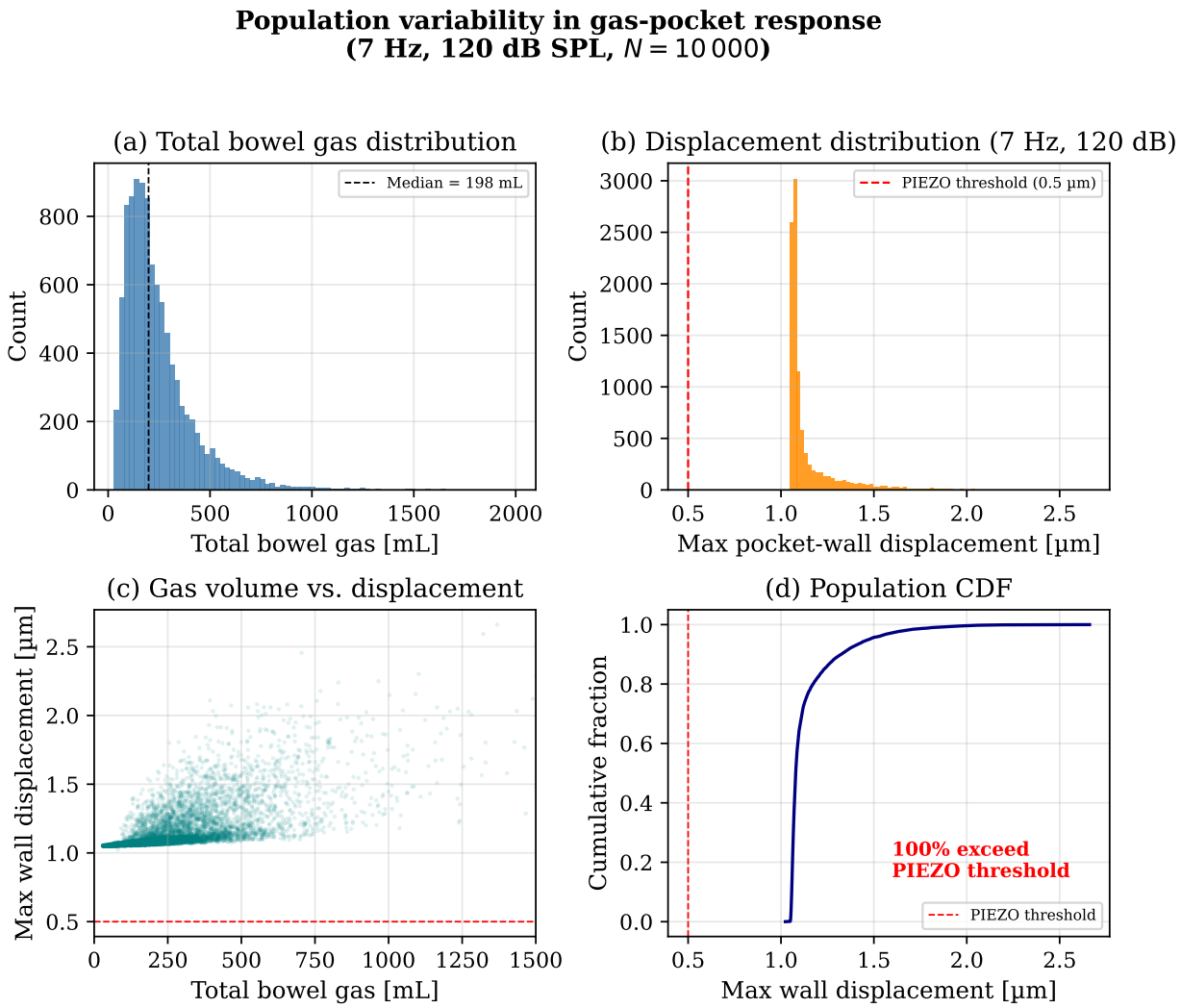

Figure 3: Population variability in gas-pocket acoustic response (7 Hz, 120 dB, $N = 10\,000$). (a) Log-normal distribution of total bowel gas. (b) Histogram of maximum pocket-wall displacement. (c) Scatter plot showing the correlation between gas volume and displacement. (d) Cumulative distribution function with the PIEZO threshold marked.

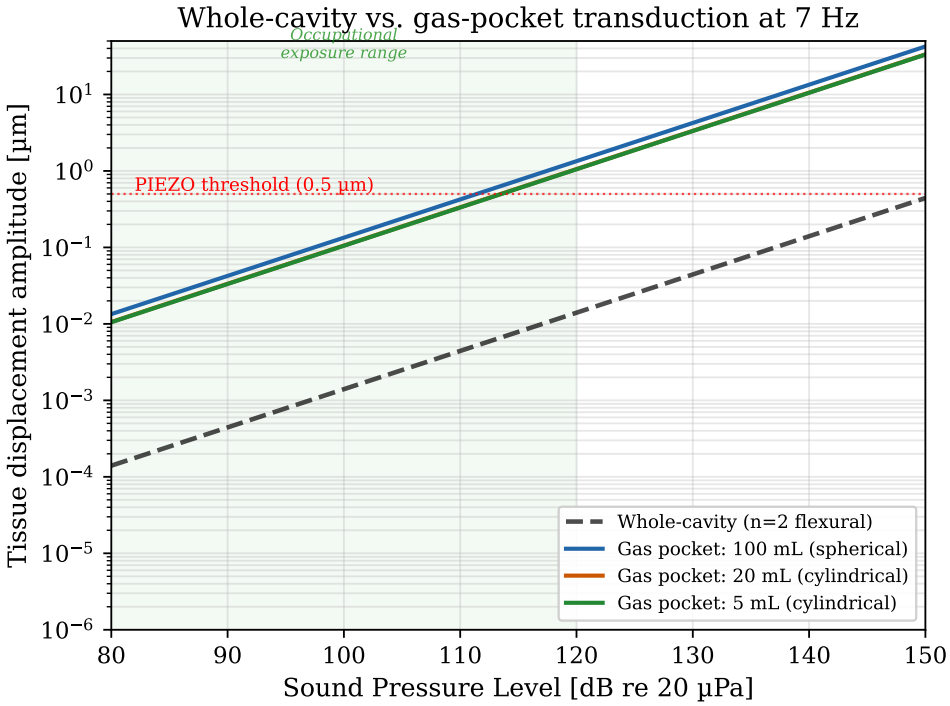

Figure 4: Tissue displacement vs. SPL at 7 Hz: gas-pocket transduction (solid lines) vs. whole-cavity flexural resonance (dashed). Gas pockets achieve the PIEZO threshold (dotted red line) at ∼111 dB to 120 dB, while the whole-cavity path requires > 180 dB.